Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Chemical Spectroscopy
Course # Chem. 340, Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1. (a) What are meant by induced absorption and stimulated emission processes? Illustrate these processes with energy diagram and example. 1.5+1.5
(b) State the characteristics of electromagnetic radiation. Write the different regions of spectrum according to the increasing frequency of the regions. 1.5+0.5
(c) Illustrate the different factors which influence the intensity of spectral lines. 3.0
(d) Sketch a diagram of a grating spectrometer. State the purposes of the use of slits in grating spectrometer. 1+1
(e) What is meant by Doppler broadening? How could you remove Doppler broadening? 4.0

2. (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 1+2
(b) Mention the limitations of microwave spectroscopy. 2.0
(c) In the rotational spectrum of CO at 25° C, the $J = 3 \rightarrow J = 4$ transition is more intense than the $J = 0 \rightarrow J = 1$ transition. Explain. 3.0
(d) Explain the reason why the rotational constants of $^{12}C^{16}O$ and $^{13}C^{16}O$ are 1.92118 cm^{-1} and 1.83669 cm^{-1}, respectively. 2.0
(e) The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the moment of inertia (I) and bond length of CO. The relative atomic weight C = 12.00 and O = 15.9994, the absolute mass of H = 1.67343×10^{-27}kg. 4.0

3. (a) How many normal modes of vibration are possible for the following molecules: HBr, O_2, CH_4 and BCl_3 2.0
(b) Write down the equation for the vibrational energy of a simple harmonic oscillator (a harmonic diatomic molecule). From this equation show that the diatomic molecule can never have zero vibrational energy. What is the selection rule for harmonic oscillator undergoing vibrational changes? 3.0
(c) Explain why does spacing between vibrational levels in a Morse curve decrease gradually? What are the zero point energies in relation to harmonic and anharmonic oscillators? 2+1
(d) Sketch the symmetry and fundamental vibrations of the carbon dioxide molecule in a figure. Explain them clearly. 2.0
(e) Explain why the C=O stretching vibration of an aldehyde gives rise to a strong absorptionin the infrared, although the absorption due to C=C vibration in an alkene is normally very weak. 2.0
(f) The equilibrium oscillation frequency for NO is 1904 cm^{-1} when examined in the IR region. Calculate the force constant of the molecule. 2.0

4. (a) State and explain the building up principle for many-electron atoms withy example. 3.0
(b) (i) Sketch the electronic energy levels and show transition between them for single electron of hydrogen atom. 1.5
(ii) State the selection rule for H-atom spectrum. Which of the following transitions are allowed for H-atom spectrum? 1.5
$2p \rightarrow 4d$, $3p \rightarrow 4s$, $2p \rightarrow 4f$, $2p \rightarrow 1s$

(iii) Sketch the lines of Lyman series observed at $\frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16}, \frac{24R}{25}, \ldots$ cm^{-1} in the spectrum of H-atom. 1.5

(c) What are the values of L, S and J for the term symbols $^4P_{3/2}$ and $^2D_{5/2}$. 2.5

(d) Discuss the transitions between ^3P and ^3D levels in the helium atom with necessary energy diagram and spectrum. Why the spectrum obtained from $^3P \rightarrow {}^3D$ transitions is referred to as compound triplet. 3+1

5. (a) State Frank-Condon principle. Discuss the different possibilities of transition for a molecule which undergo electronic transition from ground state to upper state both electronically and vibrationally. 4.0

(b) Explain why homonuclear molecules such as H_2 and N_2 do not show vibrational and rotational spectra but they exhibit electronic spectra. 3.0

(c) Write down the spectral region in which the following transitions may take place : 2.0

$\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$.

(d) What are the different processes by which an excited molecule may loss its excess energy? Discuss phosphorescence process with energy diagram. 1+2.5

(e) What are the basic differences between electronic and Raman spectroscopy? 1.5

6. (a) Deduce an expression to determine the extent of interaction between spin and magnetic field. What is Larmor frequency? 2.5+1

(b) What is meant by chemical shift? How could you distinguish between *cis*-and *trans*-isomer from ^{1}H-NMR spectra? 1+1

(c) Discuss the anisotropic effect on the chemical shift values. 3

(d) Explain why TMS is used as reference substance in NMR. 1.5

(e) What is ESR? Explain hyperfine structure with example. 4.0

7. (a) Illustrate the quantum theory of Raman effect. Mention the differences between stokes and anti-stokes radiation. 2.5+1.5

(b) What are the criteria for a molecule to be Raman active? Give examples of three molecules which exhibit Raman spectra. 2.0

(c) Write energy expression and selection rule of Raman spectra in case of symmetric top molecules. Also show that *R* branch lines appear at 4B, 6B, 8B, 10B … cm^{-1} from the exciting line. 3.0

(d) State and explain the rule of mutual exclusion with example. 3.0

(e) Discuss the advantages of Raman technique over that of Infra-Red in vibrational measurements. 2.0

8. Write informatory notes on any four of the following : 3.5 × 4

(a) Anisotropic effect on Chemical shift

(b) Mossbauer Spectroscopy

(c) The positions of E.S.R. Absorptions: the g factor

(d) Quantization of Energy

(e) Building-up principles

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3rd Year Tutorial Examination 2019 (# Chem 340)
Department of Chemistry
Jahangirnagar University
Time: 45 min; Full Marks: 20; Date: 19 December, 2019

1. State the basic criteria for a substance to be IR active with example. 1
2. Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic and an anharmonic oscillators. Show the allowed transitions and the expected spectrum. 4
3. What is the selection rule for harmonic oscillator undergoing vibrational changes? 1
4. What are P, Q and R branches in the vibration-rotation spectra? Explain their origin. 4
5. The fundamental band of CO in inert carbon tetrachloride solvent is found at 2143 cm^{-1} when examined in the IR region. Calculate the force constant of the molecule. 3
6. Explain the Stokes' and anti-stokes' lines in a rotational Raman spectrum with the help of a diagram. Discuss the relative intensities of lines in the spectrum. 4
7. The wavenumber of the incident radiation in a Raman spectrometer is 20487 cm^{-1}. What is the wavenumber of the scattered Stokes radiation for the $J = 0 \rightarrow 2$ transition of $^{14}N_2$? [Given that $B = 1.9987\ cm^{-1}$] 3

3rd Tutorial Examination 2019 (# Chem 340)
Department of Chemistry
Jahangirnagar University
Time: 45 min; Full Marks: 20; Date: 28 January, 2020

1. What are the atomic quantum numbers and discuss their significance? 4
2. What is the term symbol for $S = \frac{1}{2}$, $L = 2$? 1
3. The term symbol is $^2P_{3/2}$. What are the values of L and S? 1
4. Write down the spectral region in which the following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$. 3
5. A solution gives an absorbance of 0.454 at 359 nm in a 1.00 cm cell. The molar absorptivity is 1.25×10^4 $cm^{-1}mol^{-1}L$. What is the concentration of the solution? 2
6. Define the following terms:
   (i) relaxation time; (ii) chemical shift (iii) coupling constant 6
7. What are the basic principles of electron spin resonance spectroscopy? 3
   Discuss the term hyperfine interactions.

3rd Year Tutorial Examination 2018 (# Chem 340)
Department of Chemistry
Jahangirnagar University
Time: 25 min; Full Marks: 20; Date: 29 November, 2018

Write answer of the questions 1-5 and give an explanation supporting your answer: 5×1= 5

1. The different types of energies associated with a molecule are
a) Electronic energy, b) Vibrational energy; c) Rotational energy; d) All of the mentioned

2. During the motion, if the centre of gravity of molecule changes, the molecule possess
a) Electronic energy; b) Rotational energy; c) Translational energy; d) Vibrational energy

3. The correct order of different types of energies is
a) $E_{el} >> E_{vib} >> E_{rot} >> E_{tr}$; b) $E_{el} >> E_{rot} >> E_{vib} >> E_{tr}$; c) $E_{el} >> E_{vib} >> E_{tr} >> E_{rot}$;
d) $E_{tr} >> E_{vib} >> E_{rot} >> E_{el}$

4. The region of electromagnetic spectrum for nuclear magnetic resonance is
a) Microwave; b) Radio frequency; c) Infrared; d) UV-rays

5. Which of the following is an application of molecular spectroscopy?
a) Structural investigation; b) Basis of understanding of colors; c) Study of energetically excited reaction products; d) All of the mentioned

6. The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the rotational constant (B) and bond length of CO. The relative atomic weight (C = 12.00 and O = 15.9994, the absolute mass of H = 1.67343×10^{-27} kg. 5

7. Calculate the moment of inertia of water molecule around the axis defined by the bisector of HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm (H = 1.0079, Atomic mass unit = 1.6606×10^{-27} kg). 5

8. Calculate the energy associated with 1.00 mol of photons of green light (555 nm) ($c = 3.00 \times 10^{8}$ m s^{-1}; $h = 6.626 \times 10^{-34}$ Js). 5

4th Year Tutorial Examination 2018 (# Chem 340)

Department of Chemistry

Jahangirnagar University

Time: 45 min; Full Marks: 20; Date: 18 April, 2019

1. State the basic requirement for a substance to be IR active with example. 2
2. The fundamental frequency of vibration for $H^{35}Cl$ molecule is 2886 cm^{-1}. Calculate the force constant of the $H^{35}Cl$ molecule. Given that atomic mass of $^{35}Cl = 34.97$ amu. 4
3. How many normal modes of vibration are possible for the following molecules? HBr, O_2, OCS (linear), SO_2 (bent), BCl_3, CH_4, CH_3I and C_6H_6
4. What are the atomic quantum numbers and discuss their significance?
5. What is the term symbol for $S = \frac{1}{2}$, $L = 2$?
6. The term symbol is $^2P_{3/2}$. What are the values of L and S?
7. Write down the spectral region in which the following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$.
8. A solution gives an absorbance of 0.454 at 359 nm in a 1.00 cm cell. The molar absorptivity is 1.25×10^4 $cm^{-1}mol^{-1}L$. What is the concentration of the solution?

4th Year Tutorial Examination 2018 (# Chem 340)

Department of Chemistry

Jahangirnagar University

Time: 45 min; Full Marks: 20; Date: 18 April, 2019

1. State the basic requirement for a substance to be IR active with example. 2
2. The fundamental frequency of vibration for $H^{35}Cl$ molecule is 2886 cm^{-1}. Calculate the force constant of the $H^{35}Cl$ molecule. Given that atomic mass of $^{35}Cl = 34.97$ amu. 4
3. How many normal modes of vibration are possible for the following molecules? HBr, O_2, OCS (linear), SO_2 (bent), BCl_3, CH_4, CH_3I and C_6H_6
4. What are the atomic quantum numbers and discuss their significance?
5. What is the term symbol for $S = \frac{1}{2}$, $L = 2$?
6. The term symbol is $^2P_{3/2}$. What are the values of L and S?
7. Write down the spectral region in which the following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$.
8. A solution gives an absorbance of 0.454 at 359 nm in a 1.00 cm cell. The molar absorptivity is 1.25×10^4 $cm^{-1}mol^{-1}L$. What is the concentration of the solution?

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lection rule for rigid rotator $\Delta J = \pm 1$

lection rule for simple harmonic oscillator and
rational change, $\Delta V = \pm 1$

lection rule for anharmonic oscillator

$$\Delta V = \pm 1, \pm 2, \pm 3, \text{---}$$

ure rotational Raman spectrum এর জন্য
lection rule, $\Delta J = 0, \pm 2$

Department of Chemistry
Jahangirnagar University
First Tutorial Examination 2017
Course No.- Chem-340, Course Title: Chemical Spectroscopy
Full Marks: 20, Duration: 50 mins

Explain the induced absorption and stimulated emission processes with example and energy diagram.
Mention the names of different regions of electromagnetic spectrum.
Illustrate effusive molecular-beam method to remove line broadening.
What are the purposes of the use of slits in grating spectrometer?
Illustrate the quantization of energy.
Explain why the symmetric stretching vibration of CO_2 is IR inactive while the asymmetric stretch and bending modes are IR active.

Department of Chemistry
Jahangirnagar University
2nd Tutorial Examination 2017
Course No.- Chem-340, Full Marks: 20, Duration: 50 min.

1) State the criteria for a molecule to be microwave active. Which of the following molecules may show rotational microwave absorption spectrum: 3.0
HCl, H_2, CH_3Cl, CH_2Cl_2, Cl_2, SF_6? 3.0

2) What are the limitations of microwave spectroscopy? 5.0

3) Describe briefly the isotopic effect on the rotational energy and spectrum.

4) Discuss the working principle of a microwave oven. What are the differences between microwave oven and conventional oven? 2+2=4

5) Show that absorption spectrum of rigid diatomic molecule consists of lines at 2B, 4B, 6B, 8B, ----cm^{-1}.

6) What type of information one can get about molecules from the microwave spectroscopy?

Department of Chemistry
Jahangirnagar University
4th Tutorial Examination 2017
Course No.- Chem-340, Full Marks: 20, Duration: 50 min

(1) State and explain building up principle for many electron atoms.
(2) What are meant by H- like species? Give examples.
(3) Identify the transitions which are allowed or forbidden for H-atom spectrum:
2s→4d, 2p→4d, 3p→4s, 2p→4f, 4f→4s, and 2p→1s
(4) Find the values of L, S and J for $^2D_{3/2}$ and $^4P_{5/2}$.
(5) Discuss Hydrogen atom spectrum with energy level transitions, selection rule and spectrum.
(6) How does compound triplet spectrum arise from the transitions of ^3P to ^3D states for Helium atom? Sketch the spectrum with necessary energy diagram.

Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2018
Course Title: Chemical Spectroscopy
Course No. Chem. 340; Full Mark: 70
Time: 4 hours

(*Answer any five questions*)

1. (a) What is electromagnetic radiation? Mention its important characteristics. 3
(b) Arrange different types of electromagnetic radiation with increasing order of energy. 3
(c) Why does a molecule produce a band spectrum but an atom a line spectrum? 2
(d) Discuss important factors that contribute to intensity of spectral lines. 3
(e) Calculate the energy associated with 1.00 mol of photons of green light (555 nm) [$c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ J s]. 3

2. (a) Which of the following molecules will show a microwave rotational spectrum?
H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6
(b) Classify the following molecules on the basis of the principle moment of inertia:
HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH.
(c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines.
(d) What are the limitations of the rigid rotator model? Write down the energy expression for non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra.
(e) Calculate the moment of inertia of water molecule around the axis defined by bisector of HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg).

3. (a) Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic and an anharmonic oscillators. Show the allowed transitions and the expected spectrum.
(b) How vibrational spectrum of a diatomic molecule enables us to determine anharmonicity constant and equilibrium frequency of vibration?
(c) What are *P*, *Q* and *R* branches in the vibration-rotation spectra? Explain their origin.
(d) How can intermolecular and intramolecular hydrogen bonding be identified by IR spectroscopy?
(e) The fundamental band of CO in inert carbon tetrachloride solvent is found at 2143 cm–1 when examined in the IR region. Calculate the force constant of the molecule.

4. (a) Explain the quantum theory of Raman effect. What are the differences between stokes and anti-stokes radiation?
(b) State the criteria for a molecule to be Raman active. Give at least three examples of molecules which can give Raman spectra.
(c) Symmetric stretching mode of CO_2 molecule is Raman active whereas bending and asymmetric stretching m are observed to be Raman inactive. Explain the reasons with the plot of the variation of polarizability with displacements during vibration.
(d) State and explain the rule of mutual exclusion with example.
(e) Explain why the pure rotations of spherical top molecules are completely inactive in Raman spectrosco

5. (a) Write down the wave function expression and energy expression for H atom.
(b) What are atomic quantum numbers? Discuss their significance.
(c) What are the values of L and S for the term symbols $^3S_1$ and $^2P_{3/2}$.
(d) Explain compound doublet spectra with energy diagram and spectrum for the transition between ^2P and ^2D levels in the fine H-atom spectra.
(e) Sketch the energy levels of lithium atom in a figure showing the difference in energy of *s*, *p*, and *d* states with the same value of *n* and also show some allowed transitions in the figure.

6. (a) State Franc-Condon Principle. Discuss the different possibilities of transition for a molecule which undergoes electronic transition from ground state to upper state both electronically and vibrationally.
(b) Explain why homonuclear molucules such as H_2 and N_2 do not show vibrational and rotational spectra but they exhibit electronic spectra.
(c) What are the different processes by which excited molecules can loss its excess energy? Discuss the fluorescence process with necessary diagram.
(d) Write down the spectral region in which the following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$.
(e) What are the basic differences between electronic and Raman spectroscopy?

7. (a) Which of the following nuclei can exhibit NMR spectrum: ^{37}Cl, ^{12}C, ^{1}H, ^{13}C, ^{19}F, ^{16}O and ^{31}P?
(b) Describe an expression to determine the extent of interaction between spin and magnetic field.
(c) If a particular NMR instrument operates at 300.20MHz, calculate the magnetic field strength which is required for ^1H nucleus to resonance at this frequency. Given that plank constant (h) is $6.62 \times 10\text{-}34$ J s^{-1} and $\beta_N = 5.05 \times 10\text{-}27$ JT^{-1}.
(d) Why is TMS almost universally used as a standard reference compound for hydrogen resonances in NMR spectroscopy?
(e) What is meant by chemical shift? How could you differentiate trans- and cis- isomers with the concept of chemical shift?
(f) What are the criteria for different species to be active in ESR spectroscopy? Discuss with examples of molecules/ions/radicals which can exhibit ESR spectrum.

8. Write short notes of any four of the following:
(a) Width and intensity of spectral transitions
(b) Born-Oppenheimer approximation
(c) Chemical analysis by Raman spectroscopy
(d) Spectrum of Helium atom
(e) Circular dichroism (CD) spectroscopy
(f) Mössbauer spectroscopy

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2016
Course Title: Chemical Spectroscopy
Course No: Chem - 340
Time: 4 Hours, Total Marks: 70
Answer any five questions

6. (a) What is meant by induced absorption? Differentiate between spontaneous emission and stimulated emission.. 1+1.5
(b) State the characteristics of electromagnetic radiation. 1.5
(c) What is your concept of quantization? Explain why any form of energies of a molecule is quantized. 1+2.5
(d) How could you remove Doppler line broadening with the help of Lamb dip spectroscopic method? 4

7. (a) Sketch the spectra for the following:
(i) ^{1}H-NMR spectrum of ethanol and 2
(ii) Ultra-violet spectrum of propanone. 2
(b) What are the purposes of the use of slits in grating spectrometer? 2

2. (a) Which of the following molecules will show a microwave rotational spectrum: H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6? 2
(b) Classify the following molecules on the basis of the principle moment of inertia: HCl, CH_4, CH_3F, H_2O, OCS, BCl_3 2
(c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator. Show that the absorption spectrum of the rigid diatomic molecule is expected to consist of a series of equidistant lines. 3
(d) (i) Write down the energy expression in cm^{-1} for a non-rigid rotator. Compare the change in rotational energy levels and spectrum when passing from a rigid to a non-rigid diatomic molecule with the help of a figure. 0.5+2
(ii) explain the effect of isotope substitution on rotational spectra of a diatomic molecule such as coarbon monoxide with the help of appropriate figures and equations. 1.5
(e) The rotational spectrum of $^{79}Br^{19}F$ shows a series of equidistant lines 0.71433 cm^{-1} apart. Calculate the rotational constant B_e and hence the moment of inertia and bond length of the molecule.. 3

3. (a) State the basic requirement for a substance to be IR active with example. 1
(b) Sketch the Morse curve for the anharmonic oscillator. Give the selection rule for anharmonic oscillator. Explain fundamental, hot band, and overtone transitions with energy expressions. 1+1+3
(c) The fundamental frequency of vibration for $H^{35}Cl$ molecule is 2886 cm^{-1}. Calculate the force constant of the $H^{35}Cl$ molecule. Given that atomic mass of ^{35}Cl = 34.97 amu. 3
(d) Explain why all transitions in vibrational spectroscopy are restricted to originate from ground state.(v=0). 2
(e) What are P and R branch lines? Sketch vibrational-rotational energy levels and show the transitions between them to appear P and R branch lines in vibration-rotation spectrum. 3

4. (a) Which modes of vibration in H_2O molecule are Raman active and why? 2
(b) Consider the symmetric top molecule CH_3Cl
(i) Write down the equation for the energy levels of the molecule. 1
(ii) What are the selection rules for Raman spectra in this case? 1
(iii) write the expression for the moments of inertia (I_B and I_C) 1

their J and ΔJ values at which two series of R branch and S branch lines appear in the Raman spectrum. 2

(iv) Sketch the S branch series only in a Figure and show that the S branch series occur at $6B$, $10B$, $14B$, ---------- cm^{-1} from the exciting line. 3

(c) The wavenumber of the incident radiation in a Raman spectrometer is 20487 cm^{-1}. What is the wave number of the scattered Stokes radiation for the $J = 2 \leftarrow 0$ transition of $^{14}N_2$? [Given that $B = 1.9987\ cm^{-1}$]. 3

(d) Which of the vibrations of an AB_3 molecule are infrared or Raman active when it is (i) trigonal planar (symmetric planar), (ii) trigonal pyramidal (Symmetric pyramidal)? 2

5. (a) Get the values for L and S, for $^3S_1$, $^2P_{3/2}$. 1.5

(b) How can you get the term symbol for the following? 1+1

(i) $L = 1$, $S = 1$ and $J = 2$

(ii) $L = 2$, $S = \frac{1}{2}$ and $J = 3/2$

(c) What is meant by the 'j-splitting'? 1.5

(d) (i) Sketch the energy levels of lithium atom in a figure showing the difference in energy of s, p, and d states with the same value of n and also show some allowed transitions in the figure. 2

(ii) The doublet separation of lines in the spectral series, which is scarcely observable for hydrogen, is less than 1 cm^{-1} for the $2p$ level of lithium, about 17 cm^{-1} for sodium and over 5000 cm^{-1} for cesium' –explain the term. 1

(e) Explain compound doublet spectra with energy diagram and spectrum for the transition between ^2P and ^2D levels in the fine H-atom spectra. 3

(f) Why He^+, Be^+, B^{2+} etc., show 'hydrogen-like spectra'? 1

(g) Explain why the lines in atomic spectra of Na are doulet. 2

6. (a) What is the Born-Oppenheimer approximation of electronic spectra of diatomic molecules? Explain why all molecules can give electronic spectra. 2+1

(b) Discuss Woodward-Feiser's rule to calculate the wavelength of maximum absorption with example. 2

(c) State Franc-Condon principle. If an electronic transition takes place for a molecule at ground state both electronically and vibrationally to upper state, discuss the different possibilities of transition of that molecule. 1+3

(d) What are the different processes by which an excited molecule may loss its excess energy? Discuss phosphorescence process with energy diagram. 1+2.5

(e) What are the basic differences between electronic and Raman spectroscopy? 1.5

7. (a) Deduce the expression to determine the extent of interaction between spin and magnetic field. What is Larmor frequency? 2.5+1

(b) What is meant by chemical shift? How could you differenciate between *cis*- and *trans*-isomer form ^1H-NMR spectra? 1+1

(c) Discuss the anisotropic effect on the chemical sift values. 3

(d) Explain why TMS is used as reference substance in NMR. 1.5

(e) What type of substance can exhibit E.S.R. spectrum? Give examples. 2

(f) Sketch the E.S.R. spectrum of benzene radical. Find the total electron density for six hydrogens of this radical. 2

8. Write informatory notes on **any four of the following:** $3.5 \times 4 = 14$

(a) Width and intensity of spectral transitions

(b) Breakdown of Born-Oppenheimer approximation

(c) Spectrum of Helium atom

(d) Classical theory of Raman effects, and

(e) The positions of E.S.R absorptions.

Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2015
Course Title: Chemical Spectroscopy
Course No: Chem - 340
Time: 4 Hours, Total Marks: 70

Answer any five questions

1. (a) Explain stimulated emission process with example. Differentiate between induced absorption and stimulated emission process. 1+2
(b) What are meant by Doppler and pressure broadening? Describe effusive molecular beam method to reduce the broadning. 2+1+2
(c) Sketch a diagram of grating spectrometer. Explain why sample is placed near the exit slit rather than entrance slit in the grating spectrometer. 2.5
(d) State the factors which cause the broadening of spectral lines. 2.0
(e) What do you mean by signal to noise ratio 1.5

2. (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 2.0
(b) Mention the limitations of microwave spectroscopy. 2.0
(c) What is rotational constant (B)? Describe briefly the effect of different factors on the values of B. 4.0
(d) Explain the schematic diagram why rotational energy levels become lowered when passing from rigid to non-rigid electronic molecules. 4.0
(e) Write the [illegible] transitions among rotations 2.0
(f) energy levels for a rigid diatomic molecule

3. (a) Show that a diatomic molecule which behaves as simple harmonic oscillator can never have zero vibrational energy. State selection rule for harmonic oscillator undergoing vibrational changes. 3+1
(b) What are the different possible transitions for a diatomic molecule undergoing anharmonic oscillations? Describe the transitions with energy expressions, energy diagram and spectrum. 1+4
(c) Explain the terms P and R-branch lines. When do they appear in the vibration-rotation spectrum? What is meant by band origin? 2+2+1

4. (a) What is the condition for Raman activity? Give example. 2.0
(b) Write down the expression for pure Raman rotational energy transition for a linear molecule. 3.0
(c) What is the rule of mutual exclusion? What type of additional information can you get from Raman spectroscopy? 3.0
(d) Which type of spectroscopy would be used with the pure rotational spectrum of H_2? If the bond length of H_2 is 0.074 nm, what would be the spacing of the lines in the spectrum? 4.0
(e) CO shows two bands in IR spectrum but one in Raman spectra explain.

5. (a) What is electronic spectroscopy? Show diagrammatically various types of transitions involved in this technique.

(b) Explain the effects of polar solvents on, $n \longrightarrow \pi^*$ and $\pi \longrightarrow \pi^*$ transitions.

(c) The UV spectrum for acetone shows two peaks at $\lambda_{max} = 280$ nm, ε_{max} 15 and $\lambda_{max} = 190$ nm, ε_{max} 100. Identify the electron transitions for each. Which one of these is more intense?

(d) Calculate the absorption maximum in the UV spectrum of 2,4-hexadiene.

6. (a) State and explain building up principle for many-electron atoms with example. Write the energy expression of H-atom spectrum. Sketch the electronic energy levels and transitions between them for single electron of hydrogen atom. What is Lyman-series of H-atom.

(b) Sketch the compound doublet spectrum arising from transitions between ^2P and ^2D levels in the hydrogen atom. Why is the spectrum referred to as compound doublet?

(c) How could you determine the values of L, S and J for $^4P_{5/2}$ and $^2D_{5/2}$.

(d) What are meant by hydrogen-like species? Give examples.

Li, Na, K, Rb, Cs, B^{2+}, Be^{3+}, Al^{3+}

7. (a) Give examples of nuclei with zero spin, integral spin and half-integral spins.

(b) Derive an expression to determine the extent of interaction between spin and magnetic field.

(c) What is meant by Larmor precessional frequency? Show that the Larmor precessional frequency of a spinning nucleus in a magnetic field is just equal to the frequency separation between energy levels.

(d) Explain the terms spin-lattice and spin-spin relaxation times with examples.

(e) What is meant by chemical shift? How can you differentiate geminal isomers with the concept of chemical shift values of protons?

8. Write short notes on (any four) $3.5 \times 4 =$

(a) Anisotropic effect of chemical shift

(b) Born-Oppenheimer approximation

(c) Techniques of ESR spectroscopy

(d) Quantization of energy

(e) Chemical analysis of electronic spectroscopy

Jahangirnagar University
B.Sc. (Hons.) Part-III Final Examination 2014
Course Title: Chemical Spectroscopy
Course # Chem. 340; Full Marks: 70; Time: 4 hrs.

[Answer any Five questions]

1. (a) What are the important characteristics of electromagnetic radiation? 2
(b) Arrange the different types of electromagnetic radiation with decreasing order of energy. Discuss the differences between absorption and emission spectra. 4
(c) Why does a molecule produce a band spectrum but an atom produce line spectrum? 3
(d) What do you mean by width of a spectral line? Explain "Collision broadening" and "Doppler broadening". 3
(e) Discuss the important factors that contribute to the intensity of spectral lines.

2. (a) What are meant by induced absorption, spontaneous emission and stimulated emission? Discuss the processes with energy diagram and example. 3
(b) How could you remove Doppler broadening with the help of Lamb dip spectroscopic method.
(c) Explain why any form of energies of a molecule is quantized.
(d) State the purposes of the use of slits in grating spectrometers.
(e) A certain transition involves an energy change of 4.01×10^{-22} J. If there are 1000 molecules in the ground state, what will be the equilibrium population of the excited state at 2900 K. Assume that the energy levels are non-degenerate. 2

3. (a) State the criteria for a molecule to be microwave active. Which of the following molecules may show rotational microwave absorption spectrum: 3
Br_2, H_2, HF, CH_4, CH_3Br, CH_2Cl_2, O_2 and H_2O
(b) Differentiate between prolate and oblate symmetric top molecules. 2.5
(c) Discuss the effect of isotopic substitution on the energy levels and spectrum of a diatomic molecule with example. 4
(d) What are differences between microwave oven and conventional hot oven? 1.5
(e) The pure rotational spectrum of $^1H^{131}I$ consists of a series of lines separated by 13.10 cm^{-1}. Calculate the bond length of the molecule. 3

4. (a) What is meant by polarizability ellipsoid? Which mode of vibration in H_2O molecule are Raman active and why? 1+3
(b) Explain how Raman spectroscopy is used to determine structure of a molecule. 3
(c) Sketch the schematic diagram of a Raman spectrometer. What are the advantages of Raman technique over Infra-red spectroscopy?
(d) Explain why the symmetric stretching mode of CO_2 is Raman active whereas the antisymmetric stretching and bending modes are Raman inactive.

5. (a) Sketch a schematic diagram to represent the energy levels of a diatomic molecule regarding as a anharmonic oscillator. Show the allowed transitions and the expected spectrum?
(b) Show how vibrational spectrum of a diatomic molecule enables us to determine anharmonicity constant and equilibrium frequency of vibration.
(c) Explain why real molecules do not obey exactly the law of simple harmonic motion.
(d) How can the intermolecular and intramolecular hydrogen bonding be identified by IR spectroscopy?
(e) The fundamental mode of HCl occurs at 2886 cm^{-1}. Predict the frequency of the

6. (a) Write down the wave function expression and energy expression for H atom.
(b) Give an equation to explain the position of the spectral lines of hydrogen like atom.
(c) What are atomic quantum numbers? Discuss their significance.
(d) What are the values of L and S for the term symbol $^2P_{3/2}$.
(e) Would a spectral line be observed in the following transition? Justify your answer.
(i) 2s → 1s (ii) 2p → 2p (iii) 2p → 2s

7. (a) State Franc-Condon Principle. If an electronic transition takes place for a molecule at ground state both electronically and vibationally to upper state, discuss the different possibilities of transition of that molecule.
(b) Illustrate Woodward- Fieser's rule to calculate the wavelength of maximum absorption with example.
(c) Discuss the different processes by which an excited molecule may loss its excess energy.
(d) When does R, P and Q branch lines appear in the vibration-electronic spectrum for a diatomic molecule?

8. (a) Explain why TMS is almost universally used as a standard reference compound for hydrogen resonances in NMR spectroscopy. 1
(b) What is Lamor frequency? Discuss the mechanism of nuclear magnetic resonance. 1+2
(c) Discuss anisotropic effect on the chemical shift with example. 3
(d) Explain why it is more convenient to refer ESR absorption in terms of its 'g' factor. Why is the 'g' factor value varied to a large extent in case of ionic crystals of transition metal ions? 2+2
(e) Differentiate between chemically equivalent and magnetically equivalent nuclei with examples. 2

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...ngal University
B.Sc. (Hons.) Part-III Final Examination 2015
Course Title: Chemical Spectroscopy
Course # Chem. 340; Full Marks: 70; Time: 4 hrs.

[Answer any Five questions].

1. (a) Explain why chemical species absorb radiation of certain frequencies. 2
(b) What is a spectrum? Make a sketch of the following. 5
(i) Microwave spectrum of CO
(ii) IR spectrum of CH_3OH
(iii) Visible spectrum of C_2H_5OH
(c) Draw and label a block diagram of single beam spectrophotometer and describe each component mentioning its function. 4
(d) The wavelength associated with a photon is 500 nm. Calculate its wave number, frequency and energy. 3

2. (a) Which of the following molecules will show microwave rotational spectrum: Ars, HCl, H_2, IBr, CH_4, CH_3Cl, CH_2Cl_2, O_2 and H_2O 2
(b) Write the expression of rotational energy for a rigid diatomic molecule. Show the allowed transition between the energy levels of a rigid diatomic molecule. What is the selection rule in the case of microwave spectroscopy? 4
(c) Sketch energy levels and spectrum for rigid and non-rigid diatomic molecule. Explain why the spacing between the energy levels changes when passing from rigid to non rigid rotator. 5
(d) The rotational constant (B) for $H^{35}Cl$ is observed to be 10.5909 cm^{-1}. Calculate the value of B for $H^{37}Cl$. 4

3. (a) What is the basic requirement for a substance to be IR active? Discuss with example 2
(b) Explain why the spacing between vibrational level in a morse curve decreases gradually. 2
(c) Draw the morse curve for the anharmonic oscillator. Explain fundamental, hot band and overtone transitions showing them in the curve. Write the energy expressions for the fundamental, hot band and overtone transitions. 5
(d) Write down the vibrational energy expression of a simple harmonic oscillator. Explain why transitions between any two neighbouring vibrational levels of a diatomic molecule give rise to the same energy change. 3
(e) How many normal modes of vibration are possible for the following molecules: HBr, SO_2 (bent), BCl_3, CH_3I 3

4. (a) Why He^+, Be^{3+}, B^{4+} etc. show hydrogen like spectra? 2
(b) What are the values of L and S for $^1S_0$, $^2P_{3/2}$? 2
(c) Draw the spectrum of layman series of H-atom. Show how the ionization potential can be determined from the spectrum. 4
(d) Sketch the 'compound doublet' spectrum arising from transitions between ^2P and ^2D levels in the hydrogen atom. Explain why the spectrum is referred to as a compound doublet?
(e) State the building-up rules for many-electron atoms.

5. (a) Mention the criteria of a molecule that makes it Raman active? Which modes of vibration in H_2O molecule are Raman active and why? 2+2
(b) Explain the Stokes' and anti-stokes' lines in a rotational Raman spectrum with use of a diagram. Discuss the relative intensities of lines in the spectrum.
(c) What is the rule of mutual exclusion?

Department ...
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2011
Course Title: Chemical Spectroscopy
Course # Chem. 340
Total Marks: 70, Time: 4 hours.

[Answer *ANY FIVE* questions]

1. (a) Discuss the factors which determine the intensity of spectral lines. 4
   (b) Which of the following corresponds to the wavenumber of electromagnetic radiation that stimulates molecular vibration and rotation?
   (i) 1461 cm^{-1} (ii) 4.11 cm^{-1} (iii) 14306 cm^{-1} (iv) 43098 cm^{-1} 1+1
   Justify your answer.
   (c) What do you mean by width of spectral lines? Explain collision broadening and Doppler broadening. 3
   (d) A certain transition involves an energy change of 4.005×10^{-22} J $molecule^{-1}$. If there are 1000 molecules in the ground state, what is the approximate equilibrium population of the excited state at temperature of 290 K ? What would your answer have been if the energy change was 10 times greater? 3+2

2. (a) What is the major requirement for infra-red absorption? 2
   (b) Write down the selection rule for an anharmonic oscillator. Draw the Morse curve for the anharmonic oscillator and explain fundamentals, hot bands and overtone transitions showing them in the curve. 1+4
   (c) Explain why the spacing between vibrational levels in a Morse Curve decreases gradually. What are the zero point energies in relation to harmonic and anharmonic oscillators? 2+2
   (d) The fundamental band of CO in inert carbon tetrachloride solvent is found at 2143 cm^{-1} when examined in the IR region. Find out the force constant of the molecule. Given, $\mu(CO) = 1.14\times10^{-26}$ Kg and $C = 3.0\times10^{8}$ m s^{-1}. 3

3. (a) How many normal modes of vibration are possible for the following molecules? 3
   HBr, OCS, SO_2, C_6H_6
   (b) Sketch the vibrational energy levels and some transitions between them for a diatomic molecule undergoing harmonic oscillation. 4
   (c) How can the intermolecular and intramolecular hydrogen bonding be identified by IR spectroscopy? 2
   (d) Assume the following data for the molecule $^{1}H^{35}Cl$:
   Bond length: 127.5 pm; Bond force constant: 516.3 Nm^{-1}
   Atomic masses: $^{1}H = 1.673\cdot10^{-27}$ kg, $^{35}Cl = 58.06\cdot10^{-27}$ kg.
   Calculate the zero point energy, the energy of the fundamental vibration ν_0 and the rotational constant B. 5

Jahangirnagar University
B.Sc. (Hons.) Part-III Examination. 2012
Course Title: Chemical Spectroscopy
Course: Chem. 340
Full Marks: 70, Time: 4 hours.

[Answer ANY FIVE questions]

1. (a) What is electromagnetic radiation? Describe the characteristics of electromagnetic radiation.
(b) What are meant by absorption and emission spectra?
(c) Arrange the following radiations in order of their increasing wave number: UV radiations, radiowave, gamma rays, IR radiations, X-rays, microwave radiations.
(d) What do you mean by population of states? How does the population of states affect the lines?
(e) A particular molecule is known to undergo spectroscopic transitions between the ground sate and two excited states. (i) and (ii), its lifetime in (i) being 10^{-8} s and in (ii) about 0.1 s. Calculate the approximate uncertainty in the excited energy levels and the corresponding uncertainty in the associated radiation frequency in Hertz. Comment about the natural line widths from your calculated values of radiation frequencies.

2. (a) What substances are NMR active? Why does a nucleus in a magnetic field absorb electromagnetic radiation?
(b) What is the strength of the magnetic field used in tesla if the proton resonance occurs at 2.60 MHz (Given $\beta_N = 5.05\times10^{-27}$ JT^{-1})?
(c) Define Larmor motion. Deduce an equation for the Larmor frequency of a proton in a magnetic field.
(d) What is meant by coupling? How coupling splits NMR absorption line?
(e) Give a structure for the molecule C_4H_9Br from the following NMR data: (i) doublet δ 1.04, 6H; (ii) multiplet δ 1.95, 1H; (iii) doublet δ 3.33, 2H.

3. (a) How many normal modes of vibration are possible for the following molecules: OCS (linear), SO_2 (bent) and H_2O?
(b) Write down the equation for the vibrational energy of a simple harmonic oscillator (a harmonic diatomic molecule). From this equation show that the diatomic molecule can never have zero vibrational energy.
(c) Write an energy expression for a diatomic vibrating rotator. Show the transitions for the *P* and *R* branches between the rotational vibrational energy levels of a diatomic molecule, and the spectrum arising from them? What are the *Q*, *O* and *S* branches?
(d) Explain why the C=O stretching vibration of an aldehyde gives rise to a strong absorption in the infra-red, although the absorption due to C=C vibration in an alkene is normally very weak.
(e) The equilibrium oscillation frequency for NO is 1904 cm^{-1}. Calculate the force constant of the molecule

4. (a) What is the basic requirement of a molecule to absorb microwave radiation? Which of the following molecules will show a microwave rotational spectrum. 1+3
H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O

(b) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines 2+2

(c) Discuss the effect of isotopic substitution on the energy levels and rotational spectrum of a diatomic molecule. 4

(d) HCl has a B value of 10.593 cm^{-1} and a centrifugal distortion constant D 5.3×10^{-4} cm^{-1}. Estimate vibrational frequency of the molecule. 2

5. (a) What structural information can you get from Raman spectral analysis? 2

(b) Consider the symmetric top molecule $CHCl_3$ 1

(i) Write down the equation for the energy levels of the molecule. 1

(ii) What are the selection rules for Raman spectra in this case? 1

(iii) Write the expressions for the wavenumber ($\bar{\nu}_R$ and $\bar{\nu}_S$) equations with mentioning their J and ΔJ values at which two series of R branch and S branch lines appear in the Raman spectrum. 2

(iv) Sketch the S branch series only in a Figure and show that the S branch series occur at $6B$, $10B$, $14B$ ------ cm^{-1} from the exciting line. 3

(c) If the bond length of H_2 is 0.07417 nm, what would be the spacing of the lines in the spectrum? 3

(d) Which of the vibrations of an AB_3 molecule are infrared or Raman active when it is (i) trigonal planar (symmetric planar), (ii) trigonal pyramidal (symmetric pyramidal)? 7

6. (a) Write down the wave function expression and energy expression for H atom. 4

(b) Write the Building-up principle for many electron atoms. 4

(c) Sketch the spectrum of lithium with diagram.
The doublet separation of lines in the spectral series, which is scarcely observable for hydrogen, is less than 1 cm^{-1} for the $2p$ level of lithium, about 17 cm^{-1} for sodium, and over 5000 cm^{-1} for cesium - explain the term 6

7. (a) Define the term electronic spectroscopy. What is its absorption range? 1+1

(b) How can we get the dissociation energy of a molecule in the ground state using spectral data? 4

(c) State and explain Born-Oppenheimer approximation. 3

(d) Explain the phenomena of re-emission of energy by an excited molecule 5

8. Write short notes on any four of the following: 3.5×4

(a) Fermi resonance.

(b) Hyperfine structure of ESR absorptions

(c) Franck-Condon principle.

(d) ^{13}C NMR spectroscopy.

(e) Rule of mutual exclusion.

(f) Chemical shift.